P.4 Linear Equations in Two Variables

Name _____ Period ____ Date _____

Pre-Calculus P.4 · Student sheet

I can…

- find the slope of a line.
- write the equation of a line from a point and a slope or two points.
- write equations of parallel and perpendicular lines.

Vocabulary · fill in the missing word

Slope — the ratio of the _____ in y to the change in x

Slope-intercept form — y = _____

Point-slope form — $y - y_1$ = _____

General form — Ax + By + C = _____

Part 1 · Slope slider

Graph y = mx + 2 with a slider for m.

m	2	0	−1	20
What the line looks like				

Are y = 2x and y = −(1/2)x perpendicular? _____

Part 2 · Notes and examples (fill in each note, then work the example)

Slope: m = _____. Horizontal lines have slope 0; vertical lines have undefined slope.

Example 1

Find the slope through each pair.

a. (−2, 0), (3, 1)

b. (−1, 2), (2, 2)

c. (0, 4), (1, −1)

Point-slope form: $y - y_1$ = _____

Example 2

Write the equation of the line through (1, −2) with slope 3, in slope-intercept form.

Example 3

Write the equation of the line through (2, 3) and (−3, 6).

Parallel lines have equal slopes; **perpendicular lines** have slopes that are _____ ($m_1 m_2 = -1$).

Example 4

Write the lines through $(2, -1)$ that are (a) parallel and (b) perpendicular to $2x - 3y = 5$.

Example 5

A machine costs \$12,000 and is worth \$2,000 after 8 years. Write a linear model for its value V after t years.

Part 3 · Partner practice

Partner 1

Write the line through $(-1, 4)$ and $(3, -4)$.

Partner 2

Write the line through $(8, 3)$ perpendicular to $y = 4x - 1$.